

Pushpendra Pandey

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RESEARCH INTERESTS

Dendroclimatology; Atmospheric circulation; Climate variability and change; Mountain meteorology; Hydro-climate extremes; Climate model evaluation.

EDUCATION

Birbal Sahni Institute of Palaeosciences (BSIP)

Ph.D. in Climate Science

Apr 2022 – Present

Lucknow, India

- Thesis: *Climate Dynamics and Tree Growth Response Assessment in the Western Himalaya using Tree-rings.*

University of Lucknow

M.A. in Geography (First Class)

2018 – 2020

Lucknow, India

- Master's thesis on downscaled climate data and extreme event analysis (Himalayan region).

Pondicherry University

B.Tech. in Information Technology (First Class)

2008 – 2012

Puducherry, India

APPOINTMENTS

Birbal Sahni Institute of Palaeosciences (BSIP)

Senior Research Fellow

Apr 2022 – Present

Lucknow, India

- Dendroclimate reconstructions (FMAM); circulation diagnostics; moisture attribution workflows; model evaluation.

PUBLICATIONS

Journal articles

- [1] Pandey, P., Ashfaq, M., Lal, D., et al. (2026). Tree-ring evidence marks year 2022 as the driest spring season in nearly four centuries in the Western Himalayas. *Sci. Total Environ.*, 1013, 181315. <https://doi.org/10.1016/j.scitotenv.2025.181315>
- [2] Jain, S., Chinthala, B. D., Pandey, P., et al. (2025). Forest-Stand Structure and Treeline Dynamics of Himalayan Fir over the Last Four Centuries in Kashmir, Western Himalaya. *Dendrochronologia*, 126442. <https://doi.org/10.1016/j.dendro.2025.126442>
- [3] Lal, D., Pandey, P., Shekhar, M., et al. Tree-Ring-Based Drought Reconstruction Reveals Increased Pre-Monsoon Droughts Over the Past Two Centuries in the Lug Valley, Kullu. *Int. J. Climatol.*, e70104. <https://doi.org/10.1002/joc.70104>
- [4] Shekhar, M., Pandey, P., Singh, S., et al. (2025). Monsoon dynamics and future projections in the Himalaya: sea surface temperature, sea level pressure, and future flood risks. *Dyn. Atmos. Oceans*, 110, 101558. <https://doi.org/10.1016/j.dynatmoce.2025.101558>
- [5] Shekhar, M., Sharma, A., Pandey, P., et al. (2024). Assessing the past and future dynamics of the Asian summer monsoon: palaeomonsoon synthesis and CMIP6. *Glob. Environ. Change Adv.*, 2, 100004. <https://doi.org/10.1016/j.gecadv.2023.100004>
- [6] Balraju, W., Shekhar, M., Pandey, P., et al. (2026). Integrated dendrochemical analyses for reconstructing historical trace elemental pollution in Asia: a systematic review. *Phys. Chem. Earth*, 144(2), 104523. <https://doi.org/10.1016/j.pce.2026.104523>

Book chapter

- [1] Chinthala, B. D., Singh, A., Shekhar, M., et al. (2023). Age-Girth Stand Structure of Himalayan Fir and Growth–NDVI Relationship in Treeline Transects of Western Himalaya. In *Ecology of Himalayan Treeline Ecotone* (pp. 455–481). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-4476-5_19

Under review

- [1] Nag, D., Pandey, P., Verma, S., et al. Early Holocene latitudinal variations in Monsoon–Westerly dynamics, Northwest Himalaya. *Under review.*
- [2] Ahmed, I., Sharma, P., Lal, D., et al. Integrated multi-data approach for reconstruction of Triloknath Glacier evolution since the Little Ice Age: geomorphic controls and climate response, Western Himalaya. *Geomorphology* (under review).

METHODS & TOOLS

- **Dendro** Detrending; chronology building; EPS; nested PCA; climate-weighted PCs; site/species screening.
- **Atmos/Ocean** WD tracking & track density; EKE; 500–700 hPa GPH composites; ERA5; HadISST/OISST; SWJ diagnostics.
- **Lagrangian** FLEXPART; two-layer dynamic recycling; regime-conditioned moisture source fractions.
- **Statistics** Bayesian linear models; GAM; block cross-validation; RE/CE; EVT; end-to-end uncertainty.
- **Computing** Python (xarray, dask, pandas, cartopy), R (dplR); HPC (OLCF Frontier, Andes).

FIELDWORK

- **Lahaul–Spiti (2024)** 2700–3900 m transects; 116 cores (*Cedrus deodara*, *Pinus wallichiana*); N/E/SE aspects.
- **Kashmir & Tugnath** Chronology development across treeline and sub-alpine bands; moisture-sensitivity screening.

WORKSHOPS & TRAINING

- **ICTS-TIFR (2024)** Data Assimilation for Weather & Climate Models, Bengaluru.
- **NCAR (2024)** WRF Tutorial.
- **IUAC, New Delhi** HPC School.
- **GSI Training (2023)** Remote Sensing & GIS, Hyderabad.

EDITORIAL & REVIEW SERVICE

- **Ad-hoc Reviewer** *Earth Systems and Environment* (Springer, 2026).

SELECTED COLLABORATIONS

- **Dr. Moetassim Ashfaq** — Oak Ridge National Laboratory (HPC climate modelling; ERA5 moisture transport; WD dynamics).
- **Dr. Manmeet Singh** — Western Kentucky University (diffusion-based precipitation super-resolution; S2S prediction).
- **Dr. Mukund Palat Rao** — Columbia University (tree-ring field reconstructions; hydroclimate variability).
- **Dr. Kieran Hunt** — University of Reading (Western Disturbance tracking; synoptic circulation diagnostics).
- **Prof. Zong-Liang Yang** — University of Texas at Austin (land–atmosphere interactions; climate model evaluation).

HONORS

- **UGC-NET** All-India Rank 5 (99.99 percentile), June 2019.
- **UGC-NET** Qualified for Professorship, Dec 2018.

SERVICE & MEMBERSHIPS

- **AcSIR Science Club** Vice President (2024–Present).
- **Societies** AGU, AMS, EGU, YESS (Student/Early-Career Member).